

broadcast ARP can be specified in the file `/proc/sys/net/ipv4/neighbor/device_name/mcast_solicit`.

Local port range: If a networking application does not specify the transport port (i.e., binds to port 0), the kernel assigns one local port to the application. The port is chosen from a range that can be controlled by file `/proc/sys/net/ipv4/ip_local_port_range`.

Security

The Linux networking stack provides many ways to perform stack hardening. This helps in making the system more secure and less vulnerable to external attacks.

Capabilities: The Linux kernel provides an API, "int capable(int cap)", which can be used to verify the credentials of a running application, and if the application does not have sufficient privileges, the process can be terminated. Linux defines several capabilities like `CAP_NET_ADMIN`, `CAP_NET_RAW`, etc. Details for all the capabilities can be looked up in the `include/linux/capability.h` file. The following code snippet shows how to verify whether the application holds privileges to open raw sockets:

```
# (icapable(CAP_NET_RAW))
return -EPERM; /* application does not have privileges to open raw sockets */
```

ICMP Redirect: ICMP Redirect is sent by gateways to a host if it, due to a wrong assumption, sends a packet to the gateway destined for the local host in the same subnet. However, an attacker can forge and send an ICMP Redirect packet to a host, to divert packets for certain destinations through the attacker's host.

Therefore, sometimes it is necessary to control ICMP Redirect messages. This can be done by editing `/proc/sys/net/ipv4/conf/<device_name>/accept_redirects`. Writing 0 in this file forces the system to ignore all ICMP Redirect messages.

Secure redirect: To combat a forged ICMP redirect message and still accept a valid ICMP redirect message, Linux provides a secure redirect mechanism, which is accessed by editing the file `/proc/sys/net/ipv4/conf/<device_name>/secure_redirect`.

Writing 0 to this file forces Linux to accept the ICMP redirect message from any host, while 1 means Linux will accept the ICMP redirect message only from gateways present in the default gateway list. This prevents most of the malicious ICMP redirects and accepts only legitimate redirects.

Cache poisoning: The ARP protocol provisions for gratuitous ARP messages, by which a host itself can announce the IP and MAC address. All the hosts receiving gratuitous ARP messages update their ARP cache with the new MAC address announced.

However, this feature can be exploited by malicious hosts that announce some other host's IP address as their MAC address. Thereby, the attacker causes all other hosts in the network to forward packets destined to a legitimate host, to the attacker.

Linux provides gratuitous ARP message processing control by editing the file `/proc/sys/net/ipv4/conf/<device_name>/arp_accept`.

For example, the command `echo 0 > /proc/sys/net/ipv4/conf/<device_name>/arp_accept` forces the system to drop all gratuitous ARP messages on the given device.



Note: ARP gratuitous messages can be useful for networking systems that provide IP address redundancy over multiple interfaces. In such cases, for example, `eth0` and `eth1` having the same IP address but different MAC addresses, only one of the interfaces remains active. As soon as an active interface goes down for some reason, the standby interface becomes active and sends a gratuitous ARP message with the same IP address but a different MAC address. Therefore, all the hosts in the network update their ARP cache and start sending traffic to the `eth1` interface, thereby providing IP address redundancy.

Cache thrashing: Since the Linux kernel accepts ARP responses (including gratuitous ARP responses) to update the ARP cache, it is possible to mount an ARP cache thrashing attack where continuous ARP responses are flooded to the victim system. To prevent such attacks, the Linux kernel provides a way to specify the minimum time interval between two consecutive ARP responses to be accepted. The time interval can be specified in the file `/proc/sys/net/ipv4/neighbor/device_name/locktime`.

ICMP flood: The ICMP echo request and echo reply messages are used to verify network connectivity between two hosts. These messages do not carry any user data or useful data, and are only used to check network connectivity. However, it is possible to flood a system with ICMP echo messages causing the target system to do unnecessary processing and making the system slow down.

Linux provides a way to control such ICMP echo messages by editing the following files:

- `/proc/sys/net/ipv4/icmp_echo_ignore_all`
- `/proc/sys/net/ipv4/icmp_echo_ignore_broadcasts`

As the names suggest, the above files can be used to selectively drop either all or broadcast ICMP echo messages. Therefore, it is possible to prevent any ICMP flood attacks using the above mentioned files.

Although this article covers some of the features of administering the Linux networking stack, Linux is much richer and more flexible than indicated here, in terms of management and control.

This flexibility makes Linux one of the most sought after operating systems in the networking industry. Linux will go a long way in the years to come. 

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